2025 WINTER SEMESTER
Programming for Data Science (BCSE207P) Lab Record
Practice Program(PP)

Date:10/3/26 **Week No.:6** **Program S. No. 34**

Title	Retail Branch Sales Analytics with CSV I/O (Base R Only)

Problem Statement

- Construct a data frame from column vectors (copied from a provided table).
- Persist data to **CSV** and load it back using **base R** functions.
- Compute per-branch **Sum**, **Average**, and **Rank** (highest total = rank 1).
- Compute **Quarter-wise averages** across all branches.
- Compute **Region-wise averages** (Q1–Q4) and an **Overall_Average**.
- Save all analytics results to separate CSV output files.

Constraints: Use **base R only** (no external packages), in-memory operations only, clean and commented code.

use these values **exactly** to construct vectors and the initial data frame:

BranchID	BranchName	Region	Segment	Q1	Q2	Q3	Q4	Q5
BR001	Vellore Town	South	Grocery	120	115	130	125	130
BR002	Chennai Central	South	Electronics	140	150	145	160	145
BR003	Mumbai Andheri	West	Grocery	110	118	122	120	122
BR004	Pune Hinjewadi	West	Home	95	100	108	112	108
BR005	Delhi Karol Bagh	North	Apparel	150	142	155	158	155
BR006	Noida Sector 18	North	Electronics	132	138	136	140	136
BR007	Kolkata Park St	East	Grocery	105	107	112	115	112
BR008	Hyderabad Hitech	South	Home	128	131	134	137	134
BR009	Ahmedabad CG Rd	West	Apparel	115	120	125	129	125
BR010	Jaipur MI Road	North	Home	108	112	118	121	118

- **Input file**: `branches_input.csv`

- **Output files**:
  - `branches_by_rank.csv`
  - `quarter_wise_averages.csv`
  - `region_wise_averages.csv`

(You may optionally place them in an `output_<regNo>_csv/` folder.)

Tasks

Task 1 — Create Data Frame from the Table

1. Create vectors for: `BranchID`, `BranchName`, `Region`, `Segment`, `Q1`, `Q2`, `Q3`, `Q4`, `Q3_Sales`.
2. Combine into a data frame **`branches`** using `stringsAsFactors = FALSE`.
3. Print `str(branches)` and the first rows with `head(branches)`.

Task 2 — Write Input CSV

- Save `branches` to **`branches_input.csv`** using `write.csv(..., row.names = FALSE, na = "")`.

Task 3 — Read from CSV for Analytics

1. Read **`branches_input.csv`** into **`branches_in`** with `read.csv(stringsAsFactors = FALSE)`.
2. Validate required columns exist.
3. Coerce quarter columns (`Q1:Q4`) to numeric if needed.

Task 4 — Per-Branch Analytics (from CSV input)

1. Compute `Sum = Q1 + Q2 + Q3 + Q4`.
2. Compute `Average = Sum / 4`.
3. Compute `Rank` where **highest Sum = 1** (e.g., `rank(-Sum, ties.method="min")`).
4. Sort ascending by `Rank` (and by `BranchID` to stabilize ties).
5. Write the sorted result to **`branches_by_rank.csv`**.

Task 5 — Quarter-Wise Averages (overall)

1. Use `colMeans()` (with `na.rm=TRUE`) on the numeric columns `Q1:Q4`.
2. Convert the named vector to a 2-column data frame with headers: `Quarter`, `Average`.
3. Write to **`quarter_wise_averages.csv`**.

Task 6 — Region-Wise Averages (per quarter + overall)

1. Use `aggregate()` with `by = list(Region = branches_in$Region)` over `Q1:Q4` (mean, `na.rm=TRUE`).
2. Add `Overall_Average = rowMeans(...)` across Q1–Q4.
3. Optionally sort by `Overall_Average` (descending).

4. Write to **`region_wise_averages.csv`**.

Note: Keep everything **base R**. Avoid loops—use vectorized ops (`rowSums`, `colMeans`, `aggregate`).

Program

```
## ------------------------------------------------------------
## Retail Branch Sales Analytics — CSV Workflow (Base R only)
## ------------------------------------------------------------

## 0) File names
input_csv        <- "branches_input.csv"
branches_rank_csv <- "branches_by_rank.csv"
quarter_avg_csv   <- "quarter_wise_averages.csv"
region_avg_csv    <- "region_wise_averages.csv"

## 1) Create vectors from the table
BranchID   <-
c("BR001","BR002","BR003","BR004","BR005","BR006","BR007","BR008","
BR009","BR010")
BranchName <- c("Vellore Town","Chennai Central","Mumbai Andheri","Pune
Hinjewadi","Delhi Karol Bagh",
              "Noida Sector 18","Kolkata Park St","Hyderabad Hitech","Ahmedabad
CG Rd","Jaipur MI Road")
Region     <-
c("South","South","West","West","North","North","East","South","West","North
")
Segment    <-
c("Grocery","Electronics","Grocery","Home","Apparel","Electronics","Grocery",
"Home","Apparel","Home")

Q1 <- c(120,140,110,95,150,132,105,128,115,108)
Q2 <- c(115,150,118,100,142,138,107,131,120,112)
Q3 <- c(130,145,122,108,155,136,112,134,125,118)
Q4 <- c(125,160,120,112,158,140,115,137,129,121)
Q3_Sales <- Q3  # verification column

## 2) Create data frame and write input CSV
branches <- data.frame(
  BranchID, BranchName, Region, Segment,
  Q1, Q2, Q3, Q4, Q3_Sales,
  stringsAsFactors = FALSE
)
write.csv(branches, file = input_csv, row.names = FALSE, na = "")
cat("Input CSV written:", normalizePath(input_csv, winslash = "/"), "\n")

## 3) Read input CSV for analytics
```

```
branches_in <- read.csv(input_csv, stringsAsFactors = FALSE, check.names =
TRUE)
str(branches_in)

## Validation and typing
quarter_cols  <- c("Q1","Q2","Q3","Q4")
required_cols <- c("BranchID","BranchName","Region","Segment",
quarter_cols, "Q3_Sales")
missing <- setdiff(required_cols, names(branches_in))
if (length(missing) > 0) stop("Missing columns: ", paste(missing, collapse = ", "))
for (cn in quarter_cols) if (!is.numeric(branches_in[[cn]])) branches_in[[cn]] <-
as.numeric(branches_in[[cn]])
stopifnot(all(sapply(branches_in[quarter_cols], is.numeric)))

## 4) Per-branch analytics (Sum, Average, Rank)
branches_in$Sum     <- rowSums(branches_in[ , quarter_cols], na.rm = TRUE)
branches_in$Average <- branches_in$Sum / length(quarter_cols)
branches_in$Rank    <- rank(-branches_in$Sum, ties.method = "min")
branches_by_rank    <- branches_in[order(branches_in$Rank,
branches_in$BranchID), ]
write.csv(branches_by_rank, file = branches_rank_csv, row.names = FALSE, na
= "")
cat("Branches by rank CSV:", normalizePath(branches_rank_csv, winslash = "/"),
"\n")

## 5) Quarter-wise averages
quarter_wise_avg <- colMeans(branches_in[ , quarter_cols], na.rm = TRUE)
quarter_avg_df   <- data.frame(Quarter = names(quarter_wise_avg),
                               Average = as.numeric(quarter_wise_avg),
                               stringsAsFactors = FALSE)
write.csv(quarter_avg_df, file = quarter_avg_csv, row.names = FALSE, na = "")
cat("Quarter-wise averages CSV:", normalizePath(quarter_avg_csv, winslash =
"/"), "\n")

## 6) Region-wise averages (per quarter + overall)
region_quarter_avg <- aggregate(
  branches_in[ , quarter_cols],
  by = list(Region = branches_in$Region),
  FUN = function(x) mean(x, na.rm = TRUE)
)
region_quarter_avg$Overall_Average <- rowMeans(region_quarter_avg[ ,
quarter_cols], na.rm = TRUE)
region_quarter_avg <-
region_quarter_avg[order(-region_quarter_avg$Overall_Average,
region_quarter_avg$Region), ]
write.csv(region_quarter_avg, file = region_avg_csv, row.names = FALSE, na =
"")
```

```
cat("Region-wise averages CSV:", normalizePath(region_avg_csv, winslash =
"/"), "\n")
```

Output Snapshots

```
Input CSV written: C:/Users/sanjg/OneDrive/Documents/branches_input.csv
>
```

BranchID	BranchNa	Region	Segment	Q1	Q2	Q3	Q4	Q3_Sales
BR001	Vellore To	South	Grocery	120	115	130	125	130
BR002	Chennai C	South	Electronic	140	150	145	160	145
BR003	Mumbai Ar	West	Grocery	110	118	122	120	122
BR004	Pune Hinje	West	Home	95	100	108	112	108
BR005	Delhi Karol	North	Apparel	150	142	155	158	155
BR006	Noida Sect	North	Electronic	132	138	136	140	136
BR007	Kolkata Pa	East	Grocery	105	107	112	115	112
BR008	Hyderabac	South	Home	128	131	134	137	134
BR009	Ahmedaba	West	Apparel	115	120	125	129	125
BR010	Jaipur MI R	North	Home	108	112	118	121	118

```
> str(branches_in)
'data.frame':   10 obs. of  9 variables:
 $ BranchID  : chr  "BR001" "BR002" "BR003" "BR004" ...
 $ BranchName: chr  "Vellore Town" "Chennai Central" "Mumbai Andheri" "Pune Hinjewadi" ...
 $ Region    : chr  "South" "South" "West" "West" ...
 $ Segment   : chr  "Grocery" "Electronics" "Grocery" "Home" ...
 $ Q1        : int  120 140 110 95 150 132 105 128 115 108
 $ Q2        : int  115 150 118 100 142 138 107 131 120 112
 $ Q3        : int  130 145 122 108 155 136 112 134 125 118
 $ Q4        : int  125 160 120 112 158 140 115 137 129 121
 $ Q3_Sales  : int  130 145 122 108 155 136 112 134 125 118
>
```

```
Branches by rank CSV: C:/Users/sanjg/OneDrive/Documents/branches_by_rank.csv
>
```

BranchID	BranchNa	Region	Segment	Q1	Q2	Q3	Q4	Q3_Sales	Sum	Average	Rank
BR005	Delhi Karol	North	Apparel	150	142	155	158	155	605	151.25	1
BR002	Chennai C	South	Electronic	140	150	145	160	145	595	148.75	2
BR006	Noida Sect	North	Electronic	132	138	136	140	136	546	136.5	3
BR008	Hyderabac	South	Home	128	131	134	137	134	530	132.5	4
BR001	Vellore To	South	Grocery	120	115	130	125	130	490	122.5	5
BR009	Ahmedaba	West	Apparel	115	120	125	129	125	489	122.25	6
BR003	Mumbai Ar	West	Grocery	110	118	122	120	122	470	117.5	7
BR010	Jaipur MI R	North	Home	108	112	118	121	118	459	114.75	8
BR007	Kolkata Pa	East	Grocery	105	107	112	115	112	439	109.75	9
BR004	Pune Hinje	West	Home	95	100	108	112	108	415	103.75	10

Quarter	Average
Q1	120.3
Q2	123.3
Q3	128.5
Q4	131.7

Explanation/ Observations

1. Creation of Vectors

- Separate vectors were created for each column using the c() function in Base R.
- Character vectors were created for:
  - **BranchID**
  - **BranchName**
  - **Region**
  - **Segment**
- Numeric vectors were created for the quarterly sales values:
  - **Q1**
  - **Q2**
  - **Q3**
  - **Q4**
- A duplicate vector **Q3_Sales** was created from **Q3** for verification purposes.

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2. Creation of Data Frame and Input CSV

- All the vectors were combined into a **data frame named branches** using the data.frame() function.
- stringsAsFactors = FALSE was used to ensure text values remain **character type instead of factors**.
- The data frame was then written to a CSV file named **branches_input.csv** using the write.csv() function.

- This file acts as the **input dataset for further analytics**.

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3. Reading and Validating the CSV File

- The CSV file was read into a new data frame **branches_in** using the read.csv() function.
- str(branches_in) was used to **inspect the structure of the dataset**.
- A validation step checked whether all **required columns** exist in the dataset.
- The quarterly sales columns (**Q1, Q2, Q3, Q4**) were ensured to be **numeric type** for calculations.

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4. Branch-Level Analytics (Sum, Average, Rank)

- A new column **Sum** was created using the rowSums() function to calculate the **total sales across the four quarters for each branch**.
- Another column **Average** was created by dividing the **Sum by the number of quarters (4)**.
- The **rank() function** was used to assign ranks based on the **total sales**.
- The negative sign -branches_in$Sum ensures that **branches with higher total sales receive Rank 1**.
- ties.method = "min" ensures that branches with equal totals receive the **same minimum rank**.
- The dataset was then **sorted by rank and BranchID** and saved as **branches_by_rank.csv**.

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5. Quarter-Wise Averages

- The **average sales for each quarter** across all branches were calculated.

- The colMeans() function was used to compute the **mean value for Q1, Q2, Q3, and Q4**.
- The results were stored in a compact data frame containing:
  - **Quarter**
  - **Average Sales**
- This summary was written to **quarter_wise_averages.csv**.

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6. Region-Wise Averages

- The aggregate() function was used to calculate the **average sales for each quarter grouped by Region**.
- This produced a table showing the **average Q1, Q2, Q3, and Q4 sales for each region**.
- An additional column **Overall_Average** was calculated using rowMeans() to represent the **average sales across all quarters for each region**.
- The table was optionally **sorted by Overall_Average in descending order**.
- The final result was saved as **region_wise_averages.csv**.

References